

Part 4: Arithmetic Instructions

1. **Name the basic arithmetic instructions in 8086.** ADD, SUB, MUL, DIV, INC, DEC.
2. **How does the ADD instruction work?** `ADD dest, source` adds the source value to the destination value and stores the result in the destination.
3. **How does the SUB instruction work?** `SUB dest, source` subtracts the source value from the destination value and stores the result in the destination.
4. **What is the difference between INC and ADD?** INC adds exactly 1 and does *not* affect the Carry Flag. ADD adds any value and *does* affect the Carry Flag.
5. **Explain MUL and IMUL instructions.** MUL is for **unsigned** multiplication. IMUL is for **signed** (integer) multiplication.
6. **How does DIV work?** It divides the accumulator by the source. The Quotient is stored in AL (or AX), and the Remainder is stored in AH (or DX).
7. **How do you handle carry using ADC?** Use `ADC` (Add with Carry). It adds the two numbers plus the value of the Carry Flag (1 or 0) from the previous operation.
8. **How do you handle borrow using SBB?** Use `SBB` (Subtract with Borrow). It subtracts the source and the Carry Flag (borrow) from the destination.
9. **Which flag registers are affected by arithmetic operations?** Carry (CF), Zero (ZF), Sign (SF), Overflow (OF), Parity (PF), and Auxiliary Carry (AF).
10. **How is NEG instruction used?** `NEG` finds the 2's complement of a value (effectively multiplying by -1) to change its sign.
11. **How can you perform subtraction using addition?** You `NEG` (negate) the number you want to subtract, and then `ADD` it to the other number.
12. **Explain the difference between signed and unsigned multiplication.** Unsigned multiplication treats the numbers as raw magnitude. Signed multiplication preserves the sign bit so negative numbers are calculated correctly.
13. **How is overflow detected in arithmetic operations?** If the result is too large for the destination register (changing the sign bit incorrectly), the **Overflow Flag (OF)** is set to 1.
14. **How do you add two 16-bit numbers using 8-bit registers?** First `ADD` the lower 8 bits (AL). Then `ADC` (Add with Carry) the upper 8 bits (AH) to include any carry from the first addition.
15. **How do you perform BCD addition?** Perform a normal `ADD` instruction, and then immediately use the `DAA` (Decimal Adjust Accumulator) instruction to correct the result into valid BCD.